SIMATS ENGINEERING
ASSESSMENT TOOL 3: Application Based Questions

COURSE NAME:	Cloud Computing and Big Data Analytics	**COURSE CODE:**	CSA15

COURSE OUTCOME COVERED
CO2: Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)

Dave's Taxonomy: Manipulation (Level 2) Total Marks: 25
Question Count: 5

Code of Conduct
I (***Sandesh S/ 192521078***) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.
Signature of the student: ***Sandesh S***

Assessment Tasks

Application Based Questions

1. A company experiences unpredictable traffic spikes during seasonal sales. Explain how cloud auto-scaling and load balancing can handle this demand efficiently. How does this approach reduce downtime and improve user experience? What role do monitoring tools play in this setup? Relate your answer to a real-world e-commerce scenario.

2. An organization wants to store large volumes of backup data securely in the cloud.
   Discuss how cloud storage solutions ensure durability and data redundancy.
   How can encryption and access control protect sensitive backup data?
   Explain the cost advantages compared to traditional storage systems.
   Provide an example of a disaster recovery use case.
3. A mobile app company plans to deploy its application globally. Explain how content delivery networks (CDNs) improve performance and reduce latency. How do edge servers enhance user experience across regions? Discuss how cloud providers support global availability. Relate this to a real-time streaming or gaming application.

4. A financial institution requires strict compliance and data security in cloud usage. Explain how cloud providers support regulatory compliance and auditing. What role do logging, monitoring, and encryption play in security? How can the company ensure data sovereignty requirements are met? Give an example involving sensitive financial transactions.
5. A development team wants to adopt containerization for application deployment. Explain how containers differ from virtual machines in terms of performance and resource usage. How does orchestration (like Kubernetes) simplify deployment and scaling? Discuss portability benefits across different cloud environments.
   Provide an example of a microservices-based application.

Application Based Questions Rubric

Criteria	**Weightage**	**Level 4 - Exemplary**	**Level 3 - Proficient**	**Level 2 - Developing**	**Level 1 - Beginning**
Relevance to Application	25%	Response directly addresses the given use case with strong relevance	Mostly relevant response	Partially relevant	Weak relevance
Use of Cloud Concepts	25%	Applies appropriate concepts accurately	Mostly accurate application	Basic application only	Incorrect concept usage
Practical Insight	20%	Shows strong real-world understanding	Shows reasonable practical understanding	Limited practical insight	No practical insight
Completeness	15%	Response covers all required aspects	Covers most aspects	Covers only basic aspects	Incomplete response
Clarity	15%	Clear and concise answer	Generally clear	Some confusion in expression	Poor clarity

ANSWERS – APPLICATION BASED QUESTIONS

1. Cloud Auto-Scaling and Load Balancing During Seasonal Sales

Cloud auto-scaling adjusts the number of computing resources according to demand. When traffic increases during a seasonal sale, the cloud platform can automatically add virtual machines or application instances; when traffic decreases, it can remove excess instances. This helps the company avoid paying for unused capacity while still handling sudden increases in users.

Load balancing distributes incoming requests across multiple healthy servers instead of allowing one server to become overloaded. Health checks can identify an unavailable instance and route requests to functioning instances. Together, auto-scaling and load balancing reduce bottlenecks, improve response time, and lower the risk of downtime.

Monitoring tools track indicators such as CPU and memory utilization, request rate, response time, error rate, and server health. Thresholds or predictive rules can trigger scaling actions, while alerts help administrators investigate problems before they affect customers.

For example, during a major online retailer's festive sale, thousands of customers may browse products and place orders at the same time. Auto-scaling adds application instances and the load balancer distributes requests among them. Monitoring identifies rising traffic and latency, helping the retailer maintain a responsive checkout service and provide a better user experience.

2. Secure, Durable, and Cost-Effective Cloud Backup Storage

Cloud storage services are designed to keep backup data durable by storing copies across multiple devices and, depending on the service configuration, across separate availability zones or regions. Redundancy helps protect data if a disk, server, or facility fails. Versioning and retention policies can preserve earlier copies, while periodic restore tests confirm that backups can actually be recovered.

Encryption protects confidentiality. Data should be encrypted while being transferred to the cloud and while stored, using managed or customer-managed keys as appropriate. Access controls such as least-privilege identity permissions, multifactor authentication, and separate backup administrator accounts limit who can view, modify, or delete backups. Immutable or locked backup policies can further reduce the risk of ransomware or accidental deletion.

Compared with traditional storage, cloud backup can reduce the need to purchase and maintain physical hardware, dedicated storage rooms, power, cooling, and replacement equipment. Capacity can be increased when needed, and lower-cost archival tiers may suit backups that are rarely accessed. Costs still depend on storage volume, retrieval, data transfer, retention, and redundancy choices, so lifecycle policies should be planned carefully.

For example, if a company's on-premises server is damaged by fire or ransomware, it can restore clean backup copies from a separate cloud location. Keeping protected copies and testing recovery procedures helps the

company resume operations with less data loss and downtime.

3. CDNs, Edge Servers, and Global Application Performance

A content delivery network (CDN) uses a distributed network of edge locations to cache and deliver content closer to users. Instead of every request travelling to the application's central origin server, frequently requested items such as images, video segments, scripts, and other static files can be served from a nearby edge location. This reduces network distance, latency, and load on the origin server.

Edge servers improve user experience by delivering content from locations closer to the user and by handling repeated requests locally when cached content is available. Dynamic requests that cannot be cached may still need to reach the origin, so applications should use suitable caching rules and efficient APIs. Cloud providers support global availability through regional infrastructure, multiple availability zones, global traffic routing, health checks, and replication where configured.

For example, in a live-streaming application, viewers in different countries can receive video segments from nearby CDN edge locations rather than all downloading them from one central data centre. This can reduce buffering and startup delay. For a real-time multiplayer game, edge locations and regional game servers can reduce the distance data travels, although actual latency also depends on routing, server placement, and the player's network connection.

4. Cloud Compliance, Auditing, and Financial Data Security

Cloud providers offer security and compliance features that help financial institutions meet applicable requirements, but using a compliant cloud service does not automatically make the institution's entire system compliant. The institution must select appropriate services, configure them correctly, and follow the laws and standards that apply to its data and operations.

Auditing is supported through audit trails, identity and access records, configuration history, and reports or certifications for relevant cloud services. Centralized logging records events such as sign-ins, permission changes, data access, and administrative actions. Continuous monitoring and alerts can identify suspicious activity, unusual transaction patterns, or attempts to change security settings. Logs should be protected against alteration and retained for the required period.

Encryption should protect financial data both in transit and at rest. Strong identity controls, least-privilege permissions, multifactor authentication, key-management policies, network segmentation, and regular security reviews reduce the risk of unauthorized access. To meet data sovereignty requirements, the institution should choose approved regions for data storage and processing, verify where replicas and backups are placed, review provider and service terms, and monitor data movement.

For example, a bank processing digital payments can encrypt transaction records, restrict access to authorized payment services and staff, and log each privileged action. Alerts can flag unusual access, while audit reports

support investigations and regulatory reviews. The bank should also verify that transaction data and backups remain in legally permitted locations and that its configuration satisfies the applicable requirements.

5. Containers, Virtual Machines, and Orchestration

A virtual machine (VM) runs a guest operating system on virtualized hardware. Containers package an application and its dependencies while sharing the host operating system kernel. Because containers generally do not need a separate guest operating system for each application, they are usually smaller and start faster than VMs, allowing efficient use of CPU and memory. VMs provide a stronger boundary between separate guest operating systems, while containers still require appropriate isolation and security configuration.

Container orchestration platforms such as Kubernetes automate deployment, scheduling, service discovery, health checks, restarting failed containers, and scaling replicas. When demand rises, configured autoscaling can add replicas; when a container fails, orchestration can replace it. Rolling updates and rollback capabilities can also reduce disruption during application releases.

Containers improve portability because the same container image can run consistently in compatible development, testing, and production environments, including different cloud providers. However, portability is not completely automatic: networking, storage, identity, secrets, load balancers, and other provider-specific services may need configuration.

For example, an online shopping application may have separate microservices for product search, user accounts, inventory, payments, and order processing. Each service can be packaged in a container and deployed independently. Kubernetes can scale the product-search service during heavy traffic while keeping other services at appropriate replica counts. This supports flexible scaling and deployment, provided that service communication, security, monitoring, and persistent data are managed correctly.